

Team RAMP Hardware Product Requirements Document (PRD)

AttendX

Smart Attendance Management System

September 2026.

1. Document Information & Product Summary

Product Name	AttendX
Document Version	1.0
Solution Type	Hardware / Embedded Systems
Prepared By	Team RAMP
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Product Summary

AttendX is a standalone attendance system that provides two authentication options: fingerprint and PIN. An ESP32 controls the system using an SMF V1.7 fingerprint sensor, 4×4 keypad, 20×4 LCD, and ESP-CAM. Fingerprint is the primary attendance method, while PIN provides an alternative when fingerprint recognition fails. PIN-based attendance automatically captures the individual's image and records the authentication method on the dashboard.

Purpose of the PRD

This PRD establishes the problem, users, objectives, MVP boundaries, hardware design, control logic, safety considerations, and tests that will be used to determine whether the prototype achieves its intended purpose.

2. Problem Statement

Fingerprint attendance systems can experience recognition failures, preventing individuals from recording attendance smoothly. AttendX addresses this by providing a PIN-based alternative while retaining fingerprint authentication as the primary method. When PIN authentication is used, the ESP-CAM captures the individual's image to provide additional attendance evidence.

Assumptions to Validate

- SMF V1.7 provides reliable fingerprint recognition.
- Each individual can have multiple registered fingerprints.
- Registered users can authenticate using a personal PIN.
- ESP-CAM can capture a usable image during PIN attendance.
- Wi-Fi can synchronize attendance records and captured images.
- Local storage can retain records during network outages.
- The system can support approximately 100 users.

3. Product Goal & Objectives

Goal: To develop a reliable standalone attendance system that provides fingerprint and PIN authentication, records Check-In/Check-Out activity, captures images for PIN attendance, and provides centralized attendance information.

ID	Objective	How Completion Will Be Gauged
OBJ-01	Provide reliable fingerprint and	Both authentication methods

	PIN authentication, including PIN fallback when fingerprint recognition fails	successfully identify registered users
OBJ-02	Record and manage Check-In/Check-Out with duplicate prevention and authentication mode tracking	Correct attendance time, status, and Fingerprint/PIN mode recorded
OBJ-03	Capture and associate an image with PIN-based attendance	Image linked to the correct attendance event
OBJ-04	Automatically track attendance status, lateness, and user roles	On Time/Late status, late records, and Staff/Student classification available
OBJ-05	Provide centralized, connected and resilient attendance management	Dashboard, Wi-Fi synchronization, offline storage, history, and reporting function correctly

4. Target Users & Use Context

AttendX will be used at a designated attendance terminal. Users select Check-In or Check-Out, authenticate with a registered fingerprint, and receive status feedback through the LCD. Administrators manage users through the dashboard and protected terminal registration mode.

User	Need	Desired Outcome
Staff	Record attendance	Fast, reliable authentication
Students	Record attendance	Simple attendance process
Administrator	Register users/fingerprints	Correct user association
Management	Monitor attendance	Clear attendance visibility

5. Value Proposition & Significance

AttendX provides:

- **Reliable fingerprint attendance** — focused on improving the current recognition experience.
- **Centralized monitoring** — attendance records are available through one dashboard.
- **Automatic lateness tracking** — arrivals after 9:00 AM are identified and measured.
- **Offline resilience** — attendance events are retained during temporary Wi-Fi failure.
- **Staff and student support** — both groups can be managed within one system.
- **Reduced proxy attendance** — attendance requires the registered individual's fingerprint.

6. Proposed Solution & Operating Logic

AttendX combines an ESP32-based fingerprint terminal with a central web-based backend and database. Users are first created on the dashboard with their name, ID, and role. An administrator then registers one or more fingerprints against the user's ID through the terminal.

State	Sensor Interpretation	Pump Command
Check-In	Valid fingerprint, not checked in	Record Check-In
Check-In	Already checked in	Reject duplicate
Check-In	Time $\leq$ 9:00 AM	Present / On Time
Check-In	Time $>$ 9:00 AM	Present / Late
Check-Out	Valid active attendance	Record Check-Out
Registration	Admin authenticated	Register fingerprint
Offline	No Wi-Fi	Store event locally
Online	Connection restored	Synchronize pending events

Control-Flow Diagram

Step 1	Step 2	Step 3	Step 4	Step 5	Step 6
Select Check-In, Check-Out, or Registration.	Authenticate using fingerprint where required.	Validate the user's attendance state.	Record the transaction and timestamp.	Display the result on the LCD.	Send to the central database or store locally if offline.

Figure 1. Simplified control flow.

7. Hardware & Functional Requirements

ID	Requirement	Priority	Verification	Acceptance Summary
FR-01	User Authentication — Support fingerprint and PIN authentication, including PIN fallback when fingerprint recognition fails	Must	Functional test	Registered users can successfully authenticate using either method
FR-02	Attendance Processing — Record Check-In/Check-Out, prevent duplicate Check-In, and record the authentication method used	Must	Functional & data test	Correct time, attendance state, and Fingerprint/PIN mode are recorded
FR-03	PIN Image Capture — Automatically capture and associate an image with PIN-based attendance using the ESP-CAM	Must	Camera & data test	Usable image is linked to the correct attendance event
FR-04	User & Access Management — Allow administrators to register users, fingerprints and PINs, with protected registration access	Must	Registration & security test	User credentials are correctly linked and unauthorized registration is blocked
FR-05	Attendance Rules & Monitoring — Apply the attendance time rules and track	Must	Time & dashboard test	Attendance status and related records are correctly displayed

	late status, lateness duration, occurrences, Staff/Student roles and attendance history			
FR-06	Connectivity, Storage & Reporting — Synchronize records through Wi-Fi, store data during outages, recover automatically, and provide dashboard/Excel reporting	Must	Connectivity & reporting test	Records remain intact, synchronize after recovery, and can be viewed/exported

8. Non-Functional, Safety & Physical Requirements

Area	Requirement	Prototype Target	Verification
Reliability	Attendance should not be lost during network failure	Local storage	Offline test
Security	Registration restricted to administrators	Admin PIN	Security test
Usability	Users receive clear instructions	LCD + keypad	User test
Connectivity	Reliable Wi-Fi communication	Wi-Fi enabled terminal	Connectivity test
Data Integrity	Attendance linked to correct user/time	Central database	Database inspection
Power	Support primary and backup power	DC + battery	Power test

9. MVP Scope & Prioritization

Priority	Included
Must Have	ESP32, SMF V1.7 fingerprint sensor, ESP-CAM, 20×4 LCD, 4×4 keypad, fingerprint/PIN authentication, PIN fallback, Check-In/Check-Out, duplicate Check-In prevention, PIN image capture, user registration, admin PIN protection, attendance/lateness tracking, Staff/Student roles, Wi-Fi synchronization, offline storage, central database, dashboard

Should Have	Excel export, image viewing in attendance history, synchronization status, improved dashboard reporting
Could Have	Configurable attendance schedules, advanced attendance analytics, support for multiple attendance terminals
Out of Scope	Facial recognition/matching, RFID, staff ID fallback, buzzer, financial penalties, payroll deductions, payroll management

10. Hardware Architecture & Technical Design

Input / Subsystem	Controller / Logic
Input	SMF V1.7 fingerprint sensor, 4×4 keypad
Controller	ESP32 development board
Output	20×4 LCD
Connectivity	Wi-Fi
Power	DC adapter + battery
Local Storage	Offline attendance events
Central System	Web backend + database
Dashboard	Attendance, users, roles, lateness, reports

Figure 2. Illustrative hardware block diagram.

Illustrative Bill of Materials (BOM)

Component	Qty.	Purpose	Critical Note
ESP32 Development Board	1	Main controller	Selected
SMF V1.7 Sensor	1	Fingerprint authentication	Selected
20×4 LCD	1	Display	Selected
4×4 Keypad	1	Input/mode selection	Selected
Battery	1	Backup power	TBD
DC Adapter	1	Primary power	TBD
Power Management Circuit	TBD	Dual-power management	TBD
Local Storage	TBD	Offline records	TBD
Prototype Board/PCB	TBD	Assembly	TBD
Wires/Connectors	TBD	Interconnection	As required
Enclosure	1	Physical protection	TBD

Key Technical Decisions

- ESP32: Main embedded controller and Wi-Fi interface.
- Capacitive fingerprint sensor: Primary authentication method.

- LCD: Provides terminal status and instructions.
- Keypad: Provides mode selection and ID entry.
- Offline storage: Prevents temporary network failure from losing attendance events.
- Central database: Provides a single source for attendance records.

11. Prototype Layout / Physical Design

The prototype terminal will be arranged around the user's normal interaction flow.

Zone 1 — LCD: Positioned prominently for instructions and attendance status.

Zone 2 — Fingerprint Sensor: Easily accessible for fingerprint placement.

Zone 3 — Keypad: Positioned near the LCD for mode selection and ID entry.

Zone 4 — Power: Battery and power-management components secured within the enclosure.

The final enclosure dimensions and component arrangement will be determined during prototyping.

12. Test Plan & Acceptance Criteria

Test ID	Test	Expected Result	Evidence	Objective
T-01	Fingerprint registration	Fingerprint linked to user	Terminal demo	Validate registration
T-02	Check-In	Attendance recorded	LCD/database	Validate Check-In
T-03	Check-Out	Check-Out recorded	Database	Validate Check-Out
T-04	Duplicate Check-In	Transaction rejected	Video	Validate state logic
T-05	Late Check-In	Marked Late	Database	Validate lateness
T-06	Wi-Fi failure	Event stored locally	Test record	Validate offline mode
T-07	Wi-Fi recovery	Event synchronized	Database	Validate sync
T-08	Admin PIN	Registration protected	Demo	Validate security
T-09	Staff/Student filter	Correct users shown	Screenshot	Validate dashboard
T-10	Excel export	File generated	Excel file	Validate reporting
T-11	Battery operation	System remains functional	Video	Validate backup power
T-12	User capacity	~100 users supported	Test records	Validate capacity

13. Risks, Constraints & Assumptions

Risk / Constraint	Impact	Likelihood	Mitigation
Fingerprint recognition remains unreliable	High	Medium	Test multiple users/fingers
SMF V1.7 specifications unavailable	Medium	High	Confirm specifications during implementation
Wi-Fi outage	High	Medium	Local storage + synchronization
Incorrect fingerprint association	High	Medium	Admin-controlled registration
Power interruption	High	Medium	Battery backup
Limited 3-week timeline	High	High	Prioritize MVP

14. Development Plan & Evidence

Period	Primary Outcome	Evidence	Exit Check
Week 1	PRD and proposal	PRD, BOM, architecture	Proposal ready
Week 2	Functional prototype	Hardware, firmware, initial dashboard	Basic attendance works
Week 3	Integrated MVP	Prototype, dashboard, tests	Core requirements demonstrated
Final Preparation	Improve reliability and presentation	Secured prototype, demo procedure, backup evidence	Safe and repeatable final demonstration